Lecture-12

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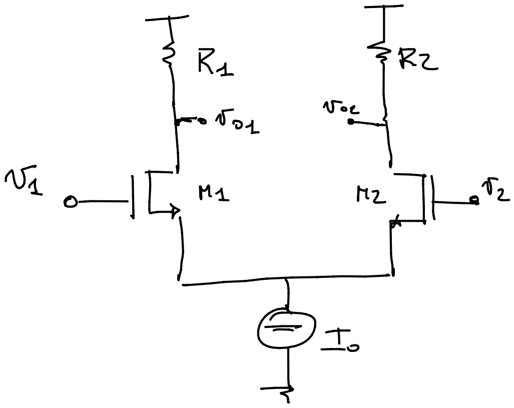
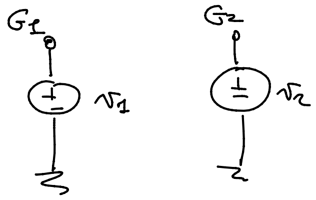
differential pair

venerdì 19 marzo 2021 11:39

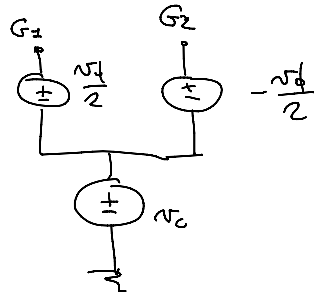


$v_1 \neq -v_2$

$$\begin{cases} v_d = v_1 - v_2 \\ v_c = \dfrac{v_1 + v_2}{2} \end{cases} \Rightarrow \begin{cases} v_1 = v_c + \dfrac{v_d}{2} \\ v_2 = v_c - \dfrac{v_d}{2} \end{cases}$$



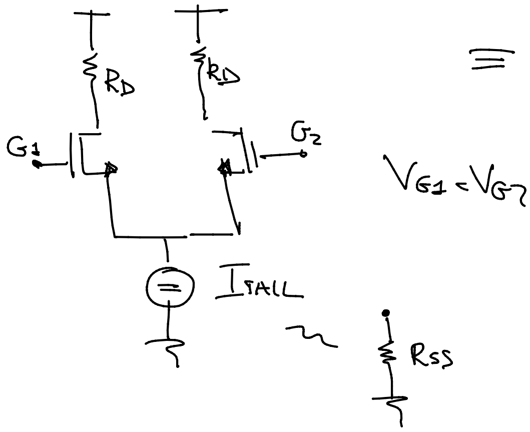
$\Rightarrow$



$$v_c = 0 \Rightarrow \begin{matrix} v_1 = \dfrac{v_d}{2} \triangleq \delta \\ v_2 = -\dfrac{v_d}{2} \triangleq -\delta \end{matrix}$$

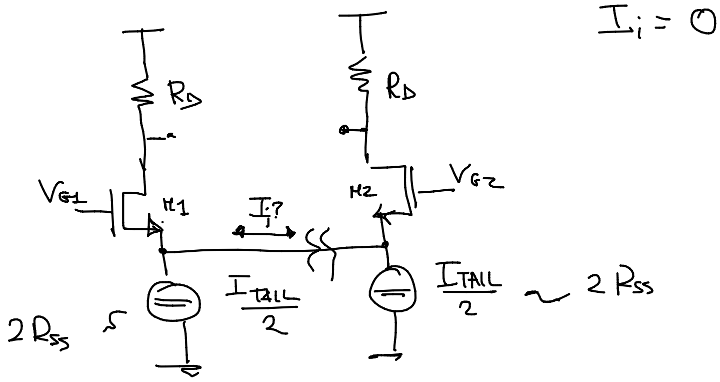
$v_c \neq 0 \qquad v_d = 0$

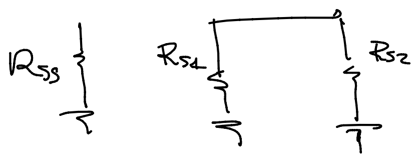
$$\begin{matrix} R_1 = R_2 \triangleq R_D \\ S_1 = S_2 \end{matrix} \Rightarrow \begin{matrix} I_1, I_2 = I \\ i_1 = i_2 \end{matrix}$$



$=$ $V_{G1} = V_{G2}$

$I_i = 0$



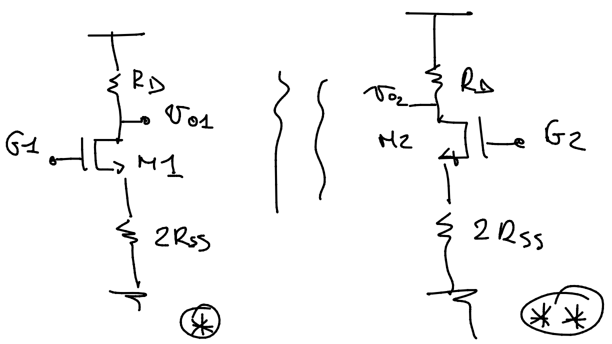


$R_{s1} // R_{s2} = R_{ss}$

$\dfrac{R_{s1} \cdot R_{s2}}{R_{s1} + R_{s2}} = R_{ss}$

$\dfrac{R_{s1}^{\cancel{2}}}{2 \cancel{R_{s1}}} = R_{ss}$

$R_{s1} = R_{s2} = 2R_{ss}$



(*) = (*)(*) = Double load

$$v_{G1} = v_{G2} = v_c$$

$$G_S = \frac{1}{2R_{SS}}$$

$$\left.\frac{v_{o1}}{v_c} = \frac{v_{o2}}{v_c} = \frac{v_o}{v_i}\right|_{DL} = -\frac{g_m}{G_D + g_o + \frac{G_T \cdot G_D}{G_S}}$$

$$G_D = \frac{1}{R_D}$$

$$= -\frac{g_m}{\frac{1}{R_D} + g_o + \frac{G_T \cdot 2R_{SS}}{R_D}} \simeq -\frac{g_m R_D}{1 + \underbrace{R_D \cdot g_o}_{\sim 0} + \underbrace{2 g_m R_{SS}}_{\text{LARGE!}}} = \frac{v_{oc}}{v_c} = A_{CM}$$

$$v_1 = \frac{v_{id}}{2} = \delta \qquad v_2 = -\frac{v_{id}}{2} = -\delta \qquad v_{S2} = v_{S1} = 0$$

$$v_{o1} = \frac{-g_{m1}}{g_{o1} + G_{L1}} \cdot \delta \qquad v_{o2} = -\frac{g_{m2}}{g_{o2} + G_L} \cdot (-\delta)$$

$$G_{L1} = G_D = \frac{1}{R_D} = G_{L2}$$

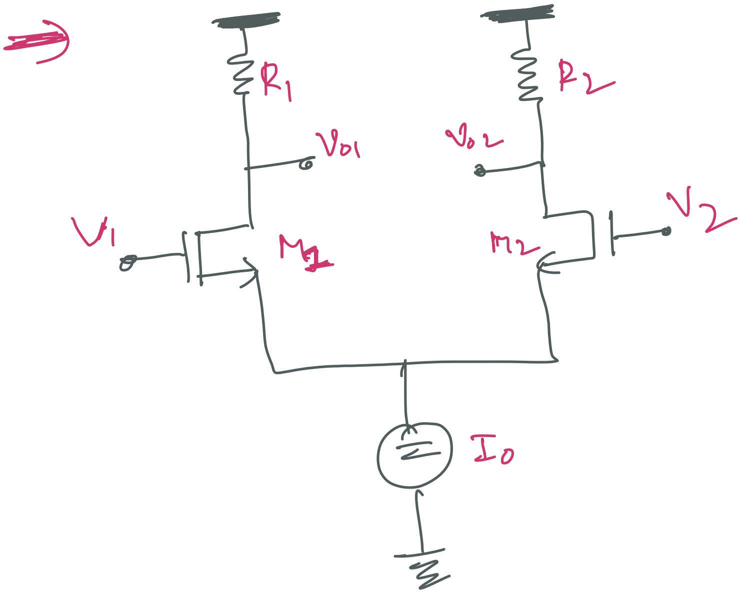
$$V_{G1o} = V_{G2o} \Rightarrow \begin{cases} g_{o1} = g_{o2} \triangleq g_o \\ g_{m1} = g_{m2} \triangleq g_m \end{cases}$$

$$v_1 - v_2 = v_{id} = 2\delta$$

$$\frac{v_{o1} - v_{o2}}{v_{id}} = -\frac{g_m}{g_o + G_D} \frac{(\cancel{2\delta})}{\cancel{2\delta}} = A_d$$

$$\underline{\underline{A_d}} \gg \underline{\underline{A_{CM}}}$$

$$v_c \neq 0$$

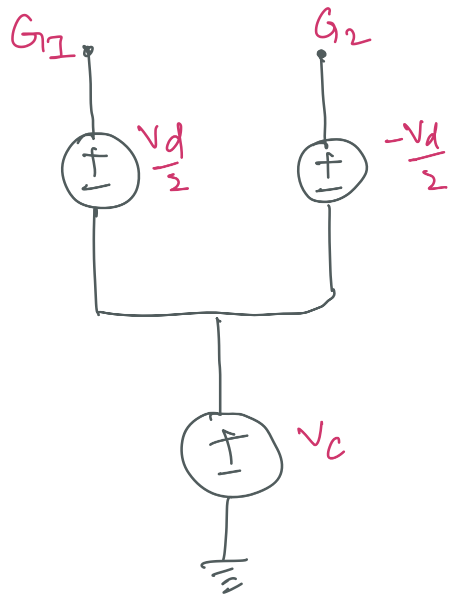


→ (V_1 can be different from $-V_2$) as we have seen in Lecture-11

To take into account all the possible instances we can take all the two input voltages as the combination of "Common Mode part" and "Differential part"

$$\boxed{\begin{aligned} V_d &= V_1 - V_2 \\ V_c &= \frac{V_1 + V_2}{2} \end{aligned}} \Rightarrow \begin{aligned} V_1 &= V_c + \frac{V_d}{2} \\ V_2 &= V_c - \frac{V_d}{2} \end{aligned}$$

These can be transformed into Basic circuit equivalents



Here G_1, G_2 are Gates of Transistor 1 & 2.

→ We have already analyzed systems where

Case-1 $V_c = 0$

i.e $V_1 = \frac{V_d}{2}$, $V_2 = -\frac{V_d}{2}$

i.e $V_1 = \delta$, $V_2 = -\delta$

→ So, we went through the analysis of the above condition in Lecture-11.

→ What we are gonna do now is to study the opposite situation.

Case 2 $V_c \neq 0$ & $V_d = 0$

If we study the Differential pair in

Small signal world where circuit is Lineare using superposition principle we can combine two cases (case 1 & case 2)

we can "Fully characterize the operation of Differential Amplifier".

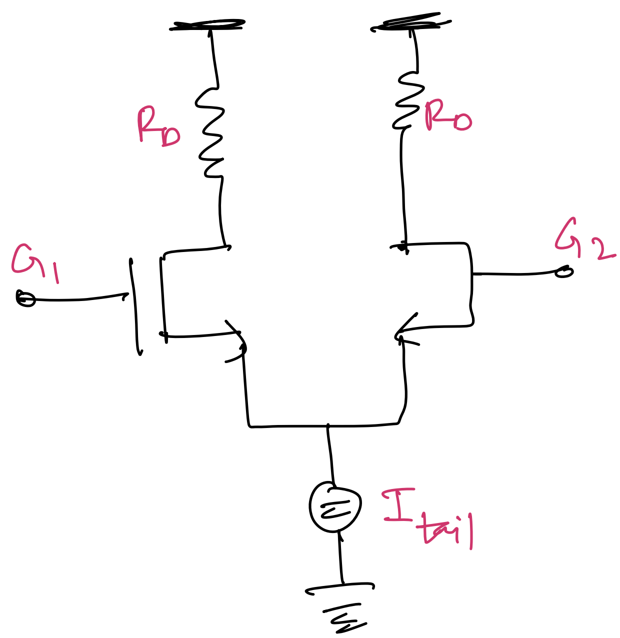
→ In order to simplify Analysis make

$R_1 = R_2 \triangleq R_D$

$S_1 = S_2$ (Source voltages are same)

→ $I_1, I_2 = I$

$i_1 = i_2$



$V_{G1} = V_{G2}$

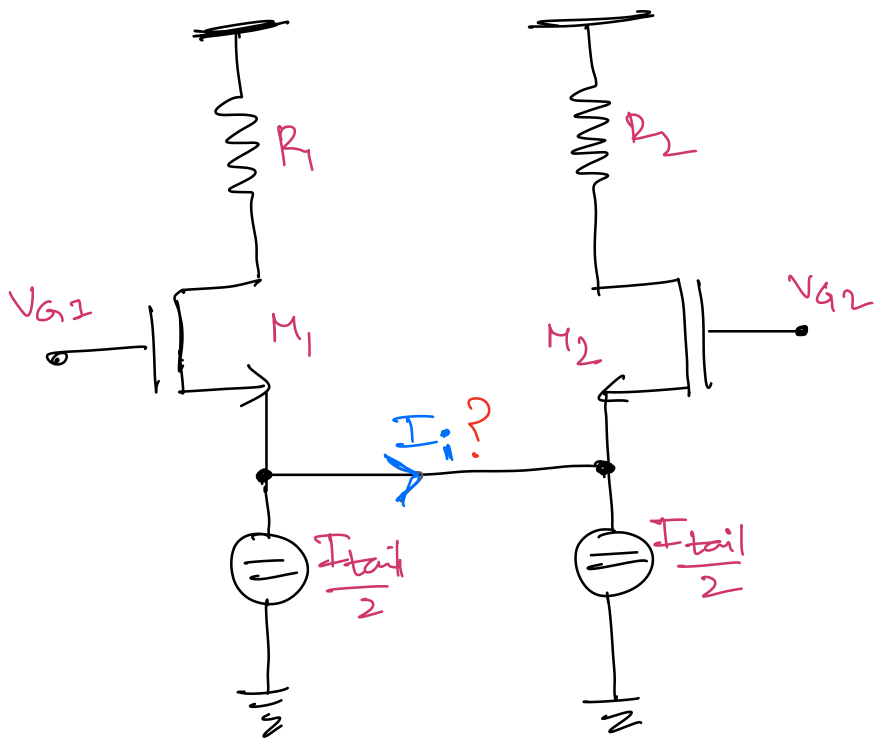
$V_{S1} = V_{S2}$

Then the overdrive voltage is also same for M_1 & M_2

So this circuit can be redrawn as

||

III



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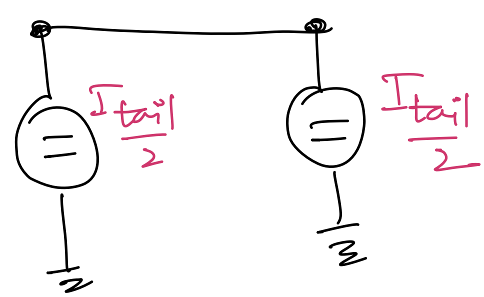
Q What would be the value of current I_i ?

By observation since both branches of the circuit are Balanced, we can make a priliminary conclusion that $I_i = 0$

Q



The Ideal current always does have a characteristic o/p Resistance associated with it.

So, if R_{SS} is the Ideal Characteristic resistance of ~~the~~ I_{tail}

Q What are the characteristic Resistance of two Ideal current sources with currents $\frac{I_{tail}}{2}$?

we should $R_{S_1} \| R_{S_2} = R_{SS}$

$$\Rightarrow \boxed{\frac{R_{S1} \cdot R_{S2}}{R_{S1} + R_{S2}} = R_{SS}}$$

let $R_{S_1} = R_{S_2}$

$$\Rightarrow \frac{R_{S_1}^{\cancel{2}}}{2 \cancel{R_{S_1}}} = R_{SS}$$

$$\Rightarrow \boxed{R_{S1} = 2R_{SS}}$$

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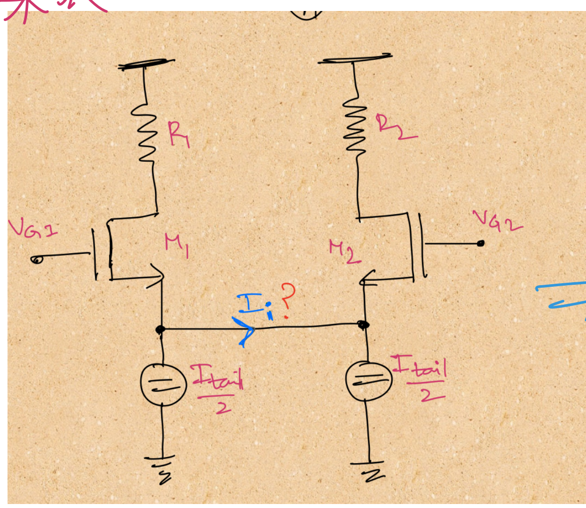
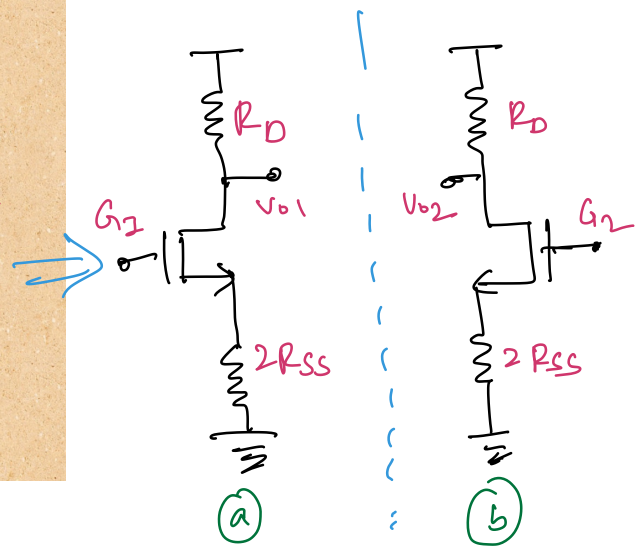




Since $I_i = 0$ we can cut the circuit

→ What are we interested to determine is the relationship b/w voltage applied to Gate and o/p V_{o1} & V_{o2}

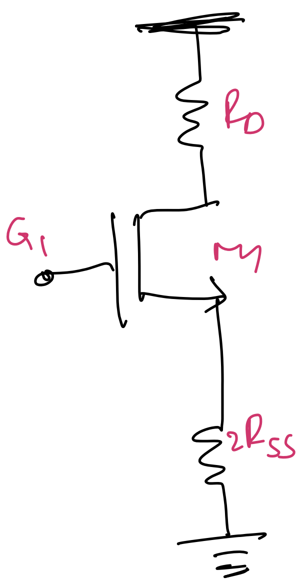
$$V_{G1} = V_{G2} = V_C \text{ (commonMode Voltage)}$$

→ Above we can see two "Double loaded circuits" after being split as $I_i = 0$

~~As~~ we have already seen Double loaded circuits in lecture-10

$$\therefore \left.\frac{V_o}{V_i}\right|_{DL} = \frac{-g_m}{G_D + g_o + G_T\left(\frac{G_D}{G_S}\right)}$$

(Double load configuration)



Here $G_S = \frac{1}{2R_{SS}}$

$G_D = \frac{1}{R_D}$

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$$\therefore \frac{V_{o1}}{V_c} = \frac{V_{o2}}{V_c} = \left.\frac{V_o}{V_i}\right|_{DL} = \frac{-g_m}{G_D + g_o + G_T\left(\frac{G_D}{G_S}\right)}$$

$$\Rightarrow \frac{-g_m}{\frac{1}{R_D} + g_o + G_T \cdot \left(\frac{2R_{SS}}{R_D}\right)}$$

$$\Rightarrow \frac{-g_m R_D}{1 + g_o R_D + G_T \cdot 2R_{ss}} \qquad G_T \simeq g_m$$

CMG

A_c

$$\Rightarrow \boxed{\frac{-g_m R_D}{1 + g_o R_D + g_m \cdot 2R_{ss}}}$$

common mode Gain.

$$\Downarrow$$

$$\Rightarrow \boxed{\frac{-g_m R_D}{1 + 2 g_m R_{ss}}}$$

common Mode Gain

** when $v_1 = \frac{V_d}{2} = \delta$, $v_2 = -\frac{V_d}{2} = -\delta$

$$\boxed{v_{o1} = \frac{-g_{m1}}{g_{o1} + G_{L1}} \cdot \delta}, \quad \boxed{v_{o2} = \frac{-g_{m2}}{g_{o2} + G_{L2}}(-\delta)}$$

In this case $v_{s1} = v_{s2} = 0$

$\therefore$ Differential pair turns out to be combination of two common source stages.

$$G_{L1} = G_D = \frac{1}{R_D} = G_{L2}$$

→ If the nominal Gate conditions

$$V_{G1o} = V_{G2o} \Rightarrow \begin{cases} g_{o1} = g_{o2} \\ g_{m1} = g_{m2} \end{cases}$$

Then we can conclude that small signal parameters are also equal.

→ This means we can simplify the formula for V_{o1} & V_{o2}

we have

$$v_1 - v_2 = v_{id}$$

$$\Rightarrow \quad v_{id} = \delta - (-\delta) = 2\delta$$

If we compute the differential o/p w.r.to to the differential input.

$$\frac{v_{o1} - v_{o2}}{v_{id}} = \frac{-g_m}{g_o + G_D} \frac{\cancel{2\delta}}{\cancel{2\delta}}$$

~~Differential Gain~~

$$\Rightarrow \boxed{\frac{v_{o1} - v_{o2}}{v_{id}} = \frac{-g_m}{g_o + G_D}} = A_d$$

Differential Gain

⇓

This is what is called (Differential gain)

→ In this expressions we are very much interested in "very large Differential Gain than common Mode Gain!"

$$\boxed{A_d \gg A_{CM}}$$

Q. How can we achieve $A_d \gg A_{cm}$?

we can play with expression

$$\boxed{A_{cm} = \frac{-g_m R_D}{1 + g_o R_D + 2 g_m R_{SS}} = \frac{v_{oc}}{v_c}}$$

$$\boxed{A_d = \frac{-g_m}{g_o + G_D} \Rightarrow -g_m R_D}$$

if g_o is neglected

∴ we can make A_{cm} very less compared to A_d by making $g_m R_{ss}$ as large as possible.

**

Differential Circuits: CMRR

This is the part that really carries information

$$v_{id} = v_{i1} - v_{i2}$$

$$v_{ic} = \frac{v_{i1} + v_{i2}}{2}$$

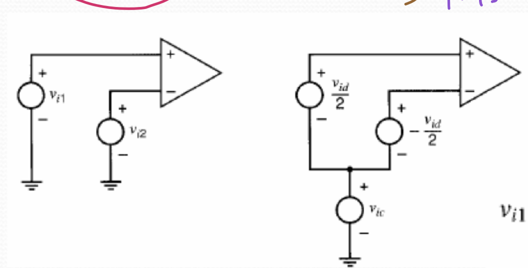
$$v_{od} = v_{o1} - v_{o2}$$

$$v_{oc} = \frac{v_{o1} + v_{o2}}{2}$$

This not so useful part.

$$v_{o1} = v_{oc} + \frac{v_{od}}{2}$$

$$v_{o2} = v_{oc} - \frac{v_{od}}{2}$$



$$v_{i1} = v_{ic} + \frac{v_{id}}{2}$$

$$v_{i2} = v_{ic} - \frac{v_{id}}{2}$$

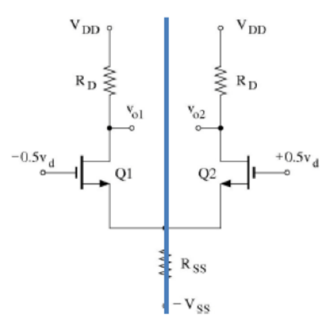
$$\boxed{\text{CMRR} = \left|\frac{A_{dm}}{A_{cm}}\right|}$$

Ratio b/w "Differential part" to "Common Mode part."

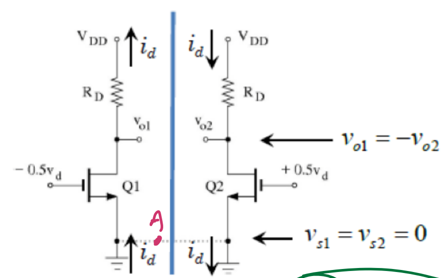
GM

**

Differential-Mode Half-Circuit



Differential Mode circuit



$v_{s1} = v_{s2} = 0$

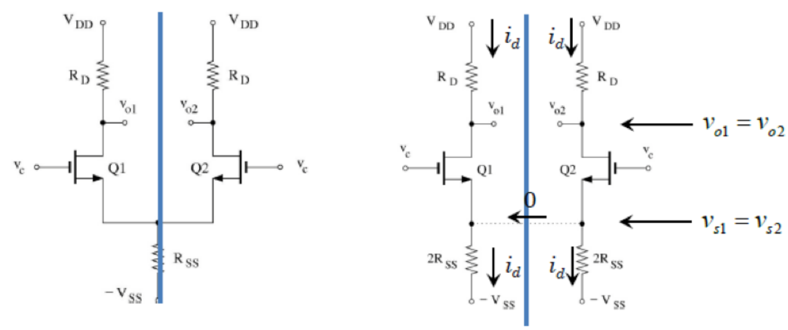
A: virtual Ground

Differential Mode Half-circuit

1. Currents about the symmetry line are equal in value and opposite in sign.
2. Voltages about the symmetry line are equal in value and opposite in sign.
3. Voltage at the summery line is zero at Node "A"

Common-Mode Half-Circuit

Common Mode circuit

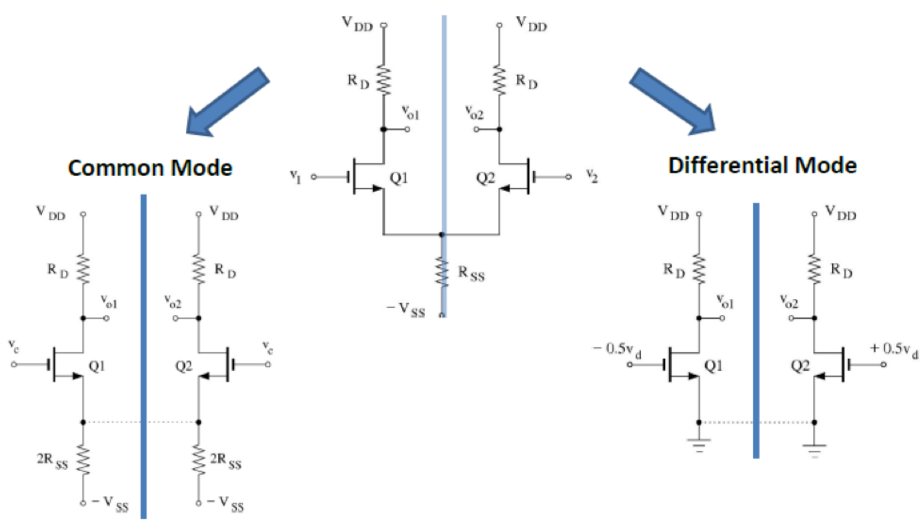


Common Mode Half-circuit

1. Currents about symmetry line are equal.
2. Voltages about the symmetry line are equal (e.g., $v_{o1} = v_{o2}$)
3. No current crosses the symmetry line.

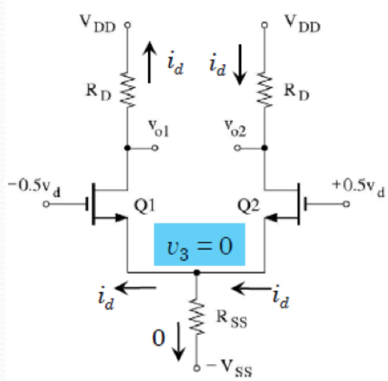
Concept of Half-Circuit

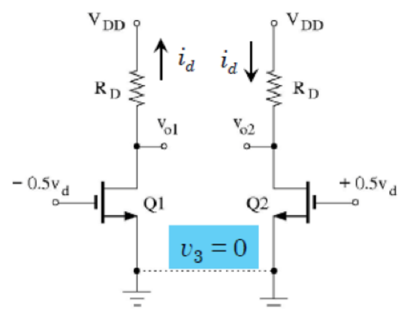
➢ **For a symmetric circuit, differential- and common-mode analysis can be performed using "half-circuits."**



Differential Amplifier- Differential Mode

$$\boxed{v_3 = 0 \quad \text{and} \quad v_{o1} = -v_{o2} \quad \Rightarrow \quad i_{d1} = -i_{d2}}$$





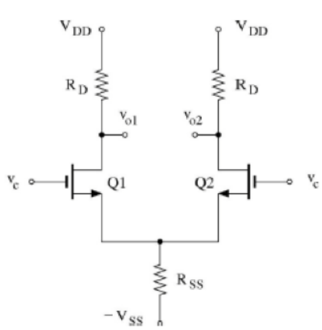
CS Amplifier

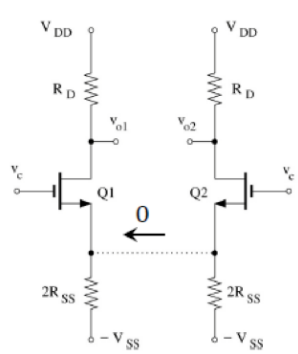
$$\frac{v_{o1}}{-0.5v_d} = -g_m(r_o \| R_D) \;, \quad \frac{v_{o2}}{+0.5v_d} = -g_m(r_o \| R_D)$$

- Because of the symmetry, the differential-mode circuit also breaks into two identical half-circuits.

Differential Amplifier- Common Mode

- The common-mode circuit breaks into two identical half-circuits.





CS Amplifiers with Rs

$$\boxed{\frac{v_{o1}}{v_c} = \frac{v_{o2}}{v_c} = -\frac{g_m R_D}{1 + 2g_m R_{SS} + R_D / r_o}}$$

Interaction of Common-Mode and Differential-Mode

$$A_{cdm} \triangleq \left.\frac{v_{od}}{v_{ic}}\right|_{v_{id}=0} \quad \text{and} \quad A_{dcm} \triangleq \left.\frac{v_{oc}}{v_{id}}\right|_{v_{ic}=0}$$

- In a perfectly balanced (symmetric) circuit, $A_{cdm} = A_{dcm} = 0$
- In practice, A_{cdm} and A_{dcm} are not zero because of component mismatch
- A_{cdm} is important because it indicates the extent to which a common-mode input will corrupt the differential output (which contains the actual signal information)

$$A_{dm} = \left.\frac{v_{od}}{v_{id}}\right|_{v_{ic}=0} = -g_m R_D$$

$$A_{cm} \simeq -\frac{g_m R_D}{1 + g_m(2R_{TAIL})} = -\frac{g_m R_D}{1 + 2g_m R_{TAIL}}$$

R_{SS}

$$\text{CMRR} \equiv \left|\frac{A_{dm}}{A_{cm}}\right| \Rightarrow \text{CMRR} = 1 + 2g_m R_{TAIL}$$

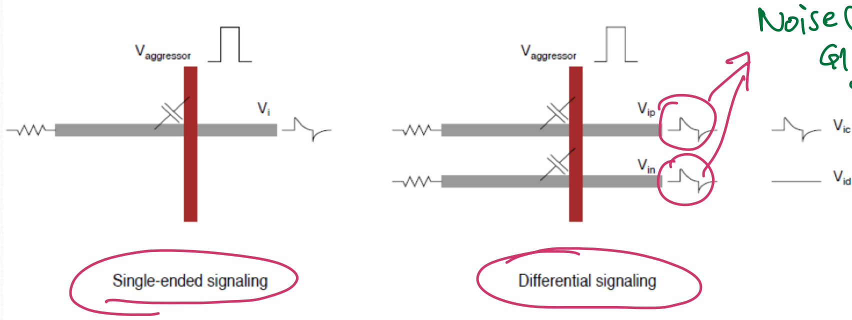
R_{SS}

CMRR should be as large as possible.

→ CMG should be minimized as much as possible to increase Noise Immunity.

Differential Pair Noise Immunity

- Differential signaling is immune to environmental noise.
- Any signal that affects both V_{in} and V_{ip} in the same way, will be ideally eliminated by the differential pair.

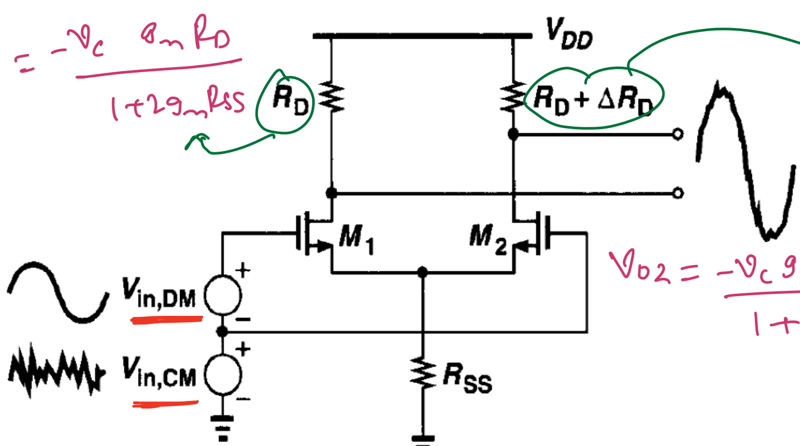


Noise (or) Glitch became of capacitive coupling

- Similar argument is applicable to other kinds of interference such as ground bounce, and supply variation.
- Mismatch between the two half-circuits, limit the achievable noise immunity.

Effect of CM noise in the presence of resistor mismatch

$$V_{o1} = \frac{-V_c \, g_m R_D}{1+2g_m R_{SS}}$$



$$V_{o2} = \frac{-V_c \, g_m (R_D+\Delta R_D)}{1+2g_m R_{SS}}$$

→ In practicle case Resistor Mismatch will became of manufacturing variations,

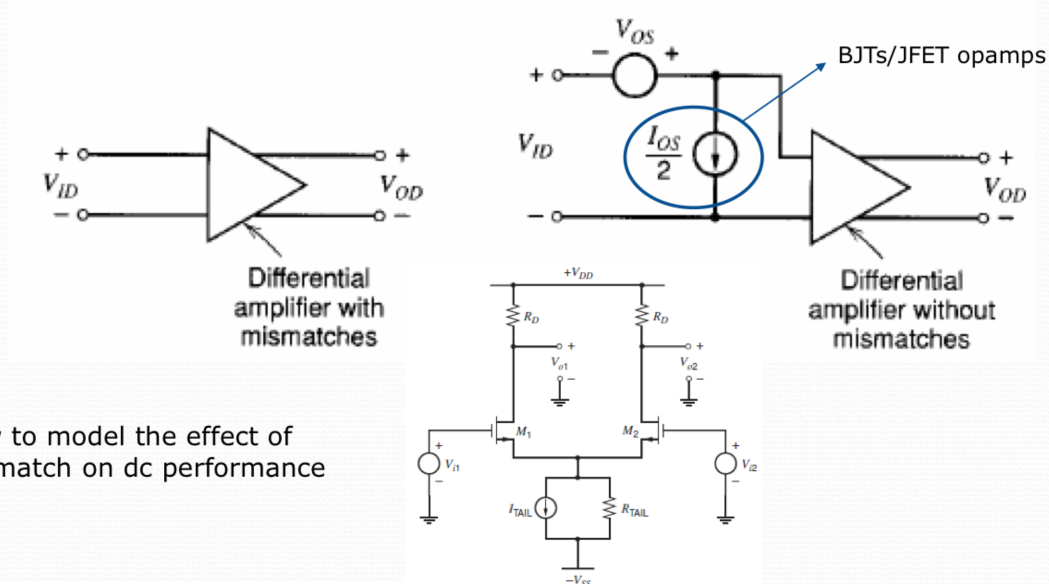
→ operating conditions also effect the Noise immunity

with perfect matching $V_{O2} - V_{O1} = 0$

with imperfect matching $V_{O2} - V_{O2} \neq 0$

→ This is fundamental issue we have in any practical Realization of circuits.

Input Offset Voltage

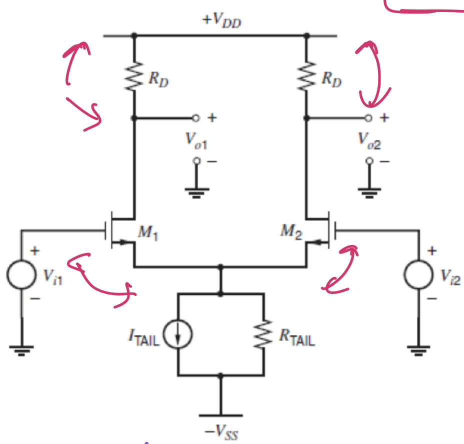


How to model the effect of mismatch on dc performance

V_{OS} is equal to the value of $V_{ID} = V_{I1} - V_{I2}$
that must be applied to the input to drive the differential output voltage $V_{OD} = V_{O1} - V_{O2}$ to zero

Input Offset-Voltage: MOSFET Pair

$V_{ID} - V_{GS1} + V_{GS2} = 0$ $\qquad V_{ID} = V_{GS1} - V_{GS2}$



$$= \boxed{V_{t1} + \sqrt{\frac{2I_{D1}}{k'(W/L)_1}} - V_{t2} - \sqrt{\frac{2I_{D2}}{k'(W/L)_2}}}$$

For V_{OD} to be zero, $I_{D1}R_{D1} = I_{D2}R_{D2}$

What could go wrong...

1 $\Delta V_t = V_{t1} - V_{t2}$

$V_t = \dfrac{V_{t1} + V_{t2}}{2}$

2 $\Delta(W/L) = (W/L)_1 - (W/L)_2$

$(W/L) = \dfrac{(W/L)_1 + (W/L)_2}{2}$

3 $\Delta R_L = R_{L1} - R_{L2}$

$R_L = \dfrac{R_{L1} + R_{L2}}{2}$

Definitions used below

$\Delta I_D = I_{D1} - I_{D2}$

$I_D = \dfrac{I_{D1} + I_{D2}}{2}$

$I_{D1} = I_D + \dfrac{\Delta I_D}{2}$ $\qquad I_{D2} = I_D - \dfrac{\Delta I_D}{2}$

$(W/L)_1 = (W/L) + \dfrac{\Delta(W/L)}{2}$ $\qquad (W/L)_2 = (W/L) - \dfrac{\Delta(W/L)}{2}$

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$\rightarrow$ $V_{ID} = V_{GS1} - V_{GS2}$ input difference voltage is equal to the difference b/w V_{GS1} & V_{GS2}

we should find what should be the values of V_{GS1} & V_{GS2} so that

$$\boxed{I_{D1}R_{D1} = I_{D2}R_{D2}}$$

because this is the condition that should be satisfied for the the $V_{O1} = V_{O2}$

What could go wrong...

(1)
$$\Delta V_t = V_{t1} - V_{t2}$$
$$V_t = \frac{V_{t1} + V_{t2}}{2}$$

(2)
$$\Delta(W/L) = (W/L)_1 - (W/L)_2$$
$$(W/L) = \frac{(W/L)_1 + (W/L)_2}{2}$$

(3)
$$\Delta R_L = R_{L1} - R_{L2}$$
$$R_L = \frac{R_{L1} + R_{L2}}{2}$$

(1) Threshold differences b/w two Transistors M_1 & M_2

(2) W/L Ratio difference b/w M_1 & M_2

(3) Resistance difference in loads R_{L1} & R_{L2}

Q: What are the implication of these variations?

$V_{G1} - V_{G2}$

$$\boxed{V_{OS} = V_{t1} - V_{t2} + \sqrt{\frac{2I_{D1}}{k'(W/L)_1}} - \sqrt{\frac{2I_{D2}}{k'(W/L)_2}}} \quad (=V_{ID})$$

$$= \Delta V_t + \sqrt{\frac{2(I_D + \Delta I_D/2)}{k'[(W/L) + \Delta(W/L)/2]}} - \sqrt{\frac{2(I_D - \Delta I_D/2)}{k'[(W/L) - \Delta(W/L)/2]}}$$

$$= \Delta V_t + (V_{GS} - V_t)\left(\sqrt{\frac{1 + \Delta I_D/2I_D}{1 + \dfrac{\Delta(W/L)}{2(W/L)}}} - \sqrt{\frac{1 - \Delta I_D/2I_D}{1 - \dfrac{\Delta(W/L)}{2(W/L)}}}\right)$$

This switching is discussed below

$$\simeq \Delta V_t + \frac{(V_{GS} - V_t)}{2}\left(\frac{1 + \Delta I_D/2I_D}{1 + \dfrac{\Delta(W/L)}{2(W/L)}} - \frac{1 - \Delta I_D/2I_D}{1 - \dfrac{\Delta(W/L)}{2(W/L)}}\right) \simeq \Delta V_t + \frac{(V_{GS} - V_t)}{2}\left(\frac{\Delta I_D}{I_D} - \frac{\Delta(W/L)}{(W/L)}\right)$$

$$\boxed{I_{D1}R_{L1} = I_{D2}R_{L2}} \Rightarrow \frac{\Delta I_D}{I_D} = -\frac{\Delta R_L}{R_L}$$

$V_{OD}=0$

$$\boxed{V_{OS} = \Delta V_t + \frac{(V_{GS} - V_t)}{2}\left(-\frac{\Delta R_L}{R_L} - \frac{\Delta(W/L)}{(W/L)}\right)}$$

GM

differential pair -offset

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$$\sqrt{\frac{2\left(I_D+\frac{\Delta I_D}{2}\right)}{\beta'\left(\frac{W}{L}+\frac{\Delta\frac{W}{L}}{2}\right)}} = \sqrt{\frac{2I_D\cdot\left(1+\frac{\Delta I_D}{I_D\cdot 2}\right)}{\beta'\frac{W}{L}\left(1+\frac{\Delta\frac{W}{L}}{2\cdot\frac{W}{L}}\right)}} = \underbrace{\sqrt{\frac{2I_D}{\beta'\frac{W}{L}}}}_{V_{OV} = V_{GS}-V_T}\cdot\frac{\sqrt{1+\frac{\Delta I_D}{2I_D}}}{\sqrt{1+\frac{\Delta\frac{W}{L}}{\frac{W}{L}\cdot 2}}}$$

Non linear

Taylor

$$\boxed{\sqrt{1+x} \simeq 1+\frac{x}{2}}$$

$k-1\rightarrow x$

$\sqrt{k}\cong\frac{k+1}{2}$ (if $k\sim 1$)

$$\Rightarrow \quad k=\frac{\left(1+\frac{\Delta I_D}{2I_D}\right)}{\left(1+\frac{\Delta\frac{W}{L}}{\frac{W}{L}\cdot 2}\right)}$$

linear

$$= \underbrace{\frac{V_{OV}}{2}\cdot\left[\frac{1+\frac{\Delta I_D}{2I_D}}{1+\frac{\Delta\frac{W}{L}}{\frac{W}{L}\cdot 2}}+1\right]}$$

we convert Non-linear to linear to analyze statistical property of the behaviours.

$$\left(\frac{1+\frac{\Delta I_D}{2I_D}}{1+\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}}-\frac{1-\frac{\Delta I_D}{2I_D}}{1-\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}}\right) = \frac{\left(1+\frac{\Delta I_D}{2I_D}\right)\left(1-\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}\right)-\left(1-\frac{\Delta I_D}{2I_D}\right)\left(1+\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}\right)}{1-\cancel{\left(\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}\right)^2}} =$$

$$= \frac{\Delta I_D}{2I_D}+\frac{\Delta I_D}{2I_D}-\frac{\Delta\frac{W}{L}}{2\frac{W}{L}}-\frac{\Delta\frac{W}{L}}{2\frac{W}{L}} = \underbrace{\frac{\Delta I_D}{I_D}-\frac{\Delta\frac{W}{L}}{\frac{W}{L}}}$$

% variation of current − % variation of form factor

when $V_{OD}=0$

$$I_{D1}\cdot R_{L1} = I_{D2}\cdot R_{L2}, \qquad V_{O1}=V_{O2}$$

$$\left(I_D+\frac{\Delta I_D}{2}\right)\left(R+\frac{\Delta R}{2}\right) = \left(I_D-\frac{\Delta I_D}{2}\right)\left(R-\frac{\Delta R}{2}\right)$$

$$\cancel{I_D\cdot R}+\frac{\Delta I_D}{2}R+\frac{\Delta R\cdot I_D}{2} = \cancel{I_D\cdot R}-\frac{\Delta R}{2}I_D-\frac{\Delta I_D\cdot R}{2} \Rightarrow \Delta I_D\cdot R = -\Delta R\cdot I_D$$

$$\frac{\Delta I_D}{I_D} = -\frac{\Delta R}{R}$$

% variation of current = − % variation in resistance

***

controllable

$$V_{OS} = \Delta V_t + \frac{(V_{GS} - V_t)}{2}\left(-\frac{\Delta R_L}{R_L} - \frac{\Delta (W/L)}{(W/L)}\right)$$

undesired offset voltage difference

overdrive

uncontrollable

→ All these are inherent to manufacturing as designers we cannot control Intrinsic manufacturing variations.

→ But we can statistically Analyze these variations in Threshold, Resistance and Form factor

→ But what is under Designers control is overDrive voltage

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